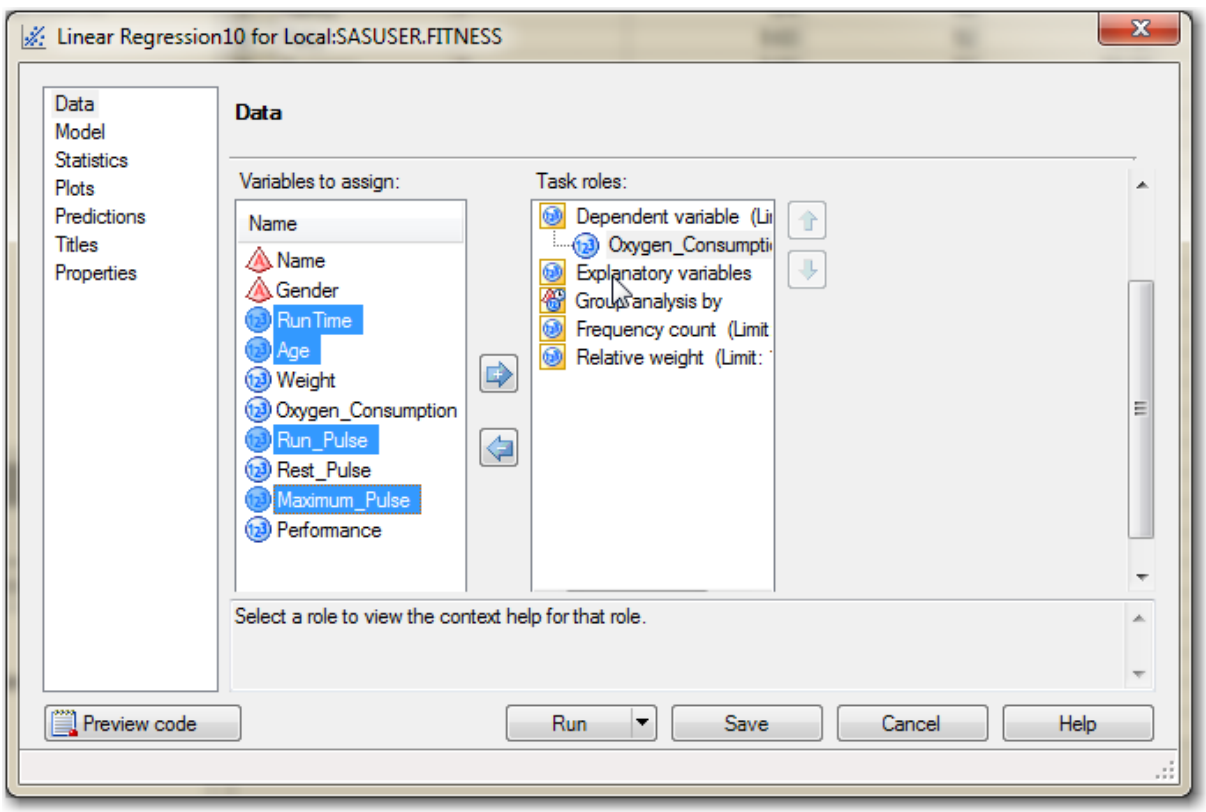
This demo contains:

- **Residual Plots**
- **Looking for Influential Observations**
- **Example of Collinearity**
- **Collinearity Diagnostics**

Residual Plots

Example: Using the **FITNESS** data set, invoke the Linear Regression task to test the regression model of **Oxygen_Consumption** against the predictor variables of **RunTime**, **Age**, **Run_Pulse**, and **Maximum_Pulse** (the model that was best based on Mallows' C_p prediction criterion). Produce the default graphics.

1. Create a new process flow and rename it **Regression Diagnostics**.
2. Open the **FITNESS** data set.
3. Select **Analyze** ⇨ **Regression** ⇨ **Linear Regression**.
4. Drag **Oxygen_Consumption** to the dependent variable task role and **RunTime**, **Age**, **Run_Pulse**, and **Maximum_Pulse** to the explanatory variables task role.



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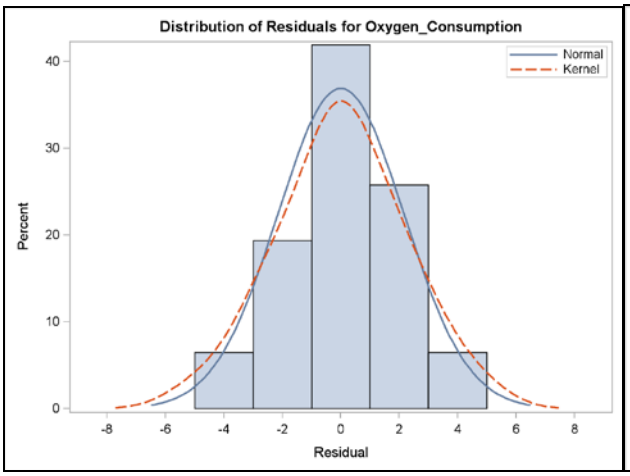
5. Click **Run**.

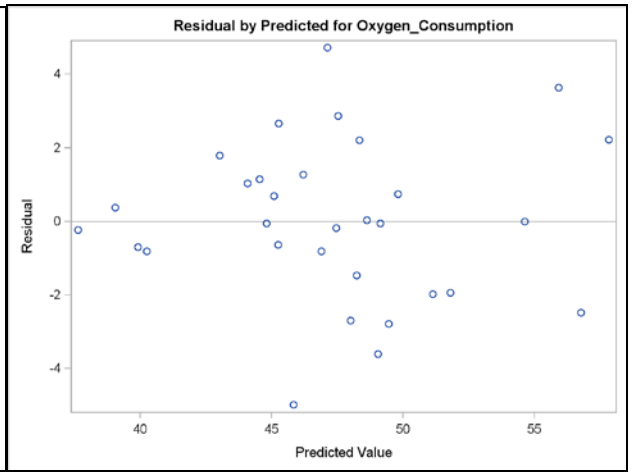
Number of Observations Read	31
Number of Observations Used	31

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	4	711.45087	177.86272	33.01	<.0001
Error	26	140.10368	5.38860		
Corrected Total	30	851.55455			

Root MSE	2.32134	**R-Square**	0.8355
Dependent Mean	47.37581	**Adj R-Sq**	0.8102
Coeff Var	4.89984		

Parameter Estimates					
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**
Intercept	1	97.16952	11.65703	8.34	<.0001
RunTime	1	-2.77576	0.34159	-8.13	<.0001
Age	1	-0.18903	0.09439	-2.00	0.0557
Run_Pulse	1	-0.34568	0.11820	-2.92	0.0071
Maximum_Pulse	1	0.27188	0.13438	2.02	0.0534





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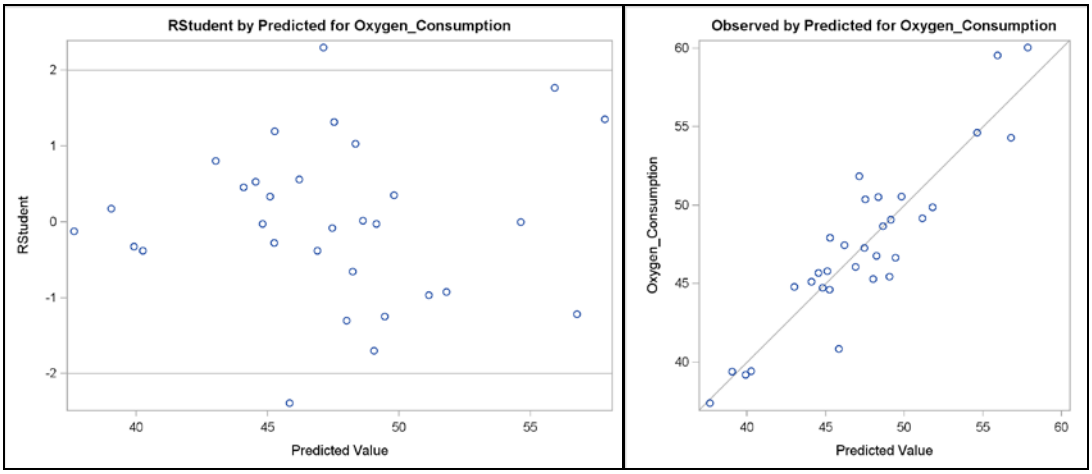
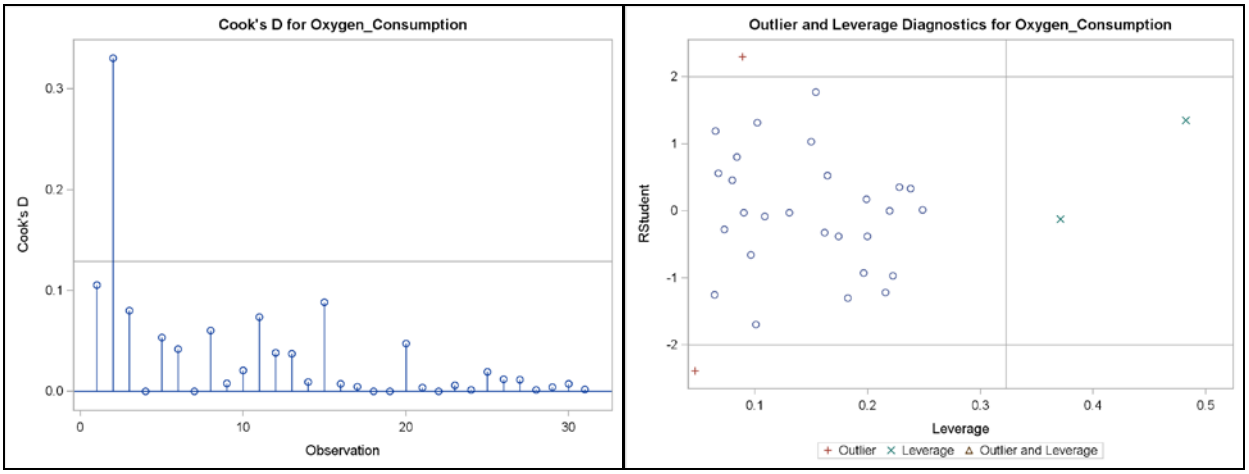
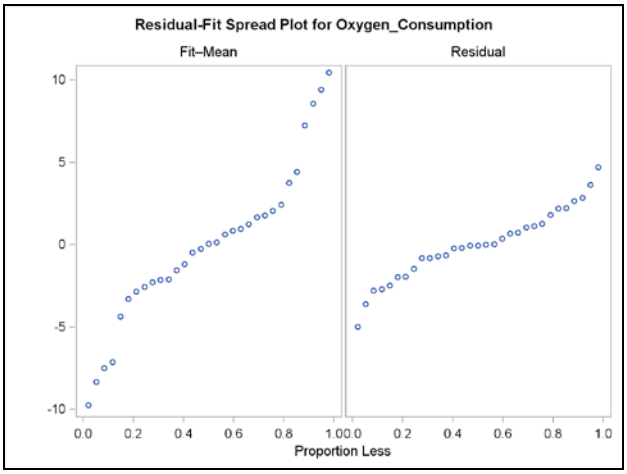
The histogram of residuals helps you to find outliers and to assess the normality assumption.







The diagnostic plots shown above are described later in greater detail.

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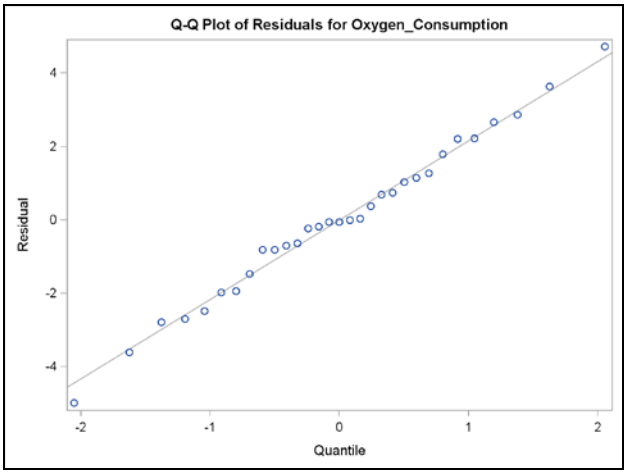
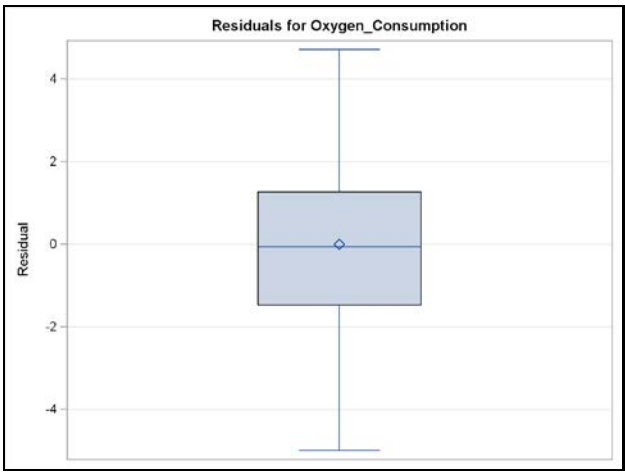
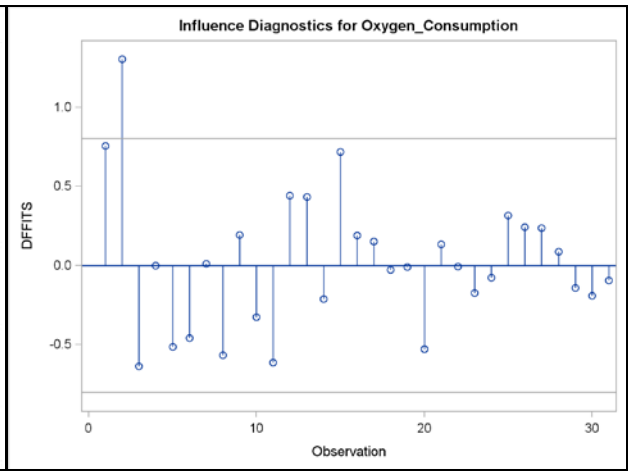
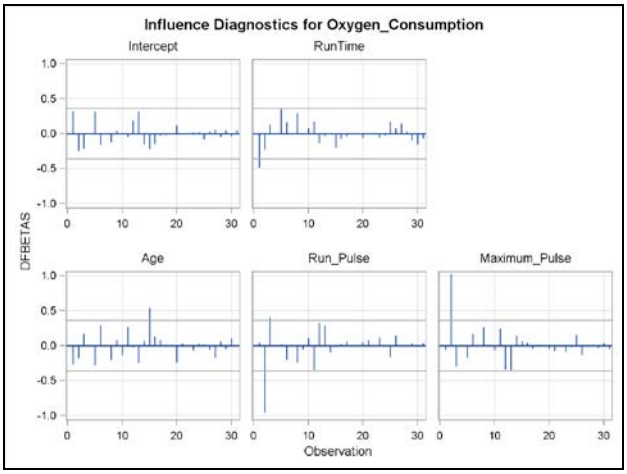


The plot of the residuals against the normal quantiles is shown above (quantile-quantile plot, also known as the Q-Q Plot). If the residuals are normally distributed, the plot should appear to follow closely a straight, diagonal line. If the plot deviates substantially from the reference line, then there is evidence against normality.

The plot shows little deviation from the expected pattern. Thus, you can conclude that the residuals do not significantly violate the normality assumption. If the residuals did violate the normality assumption, then a transformation of the response variable or a different model might be warranted.







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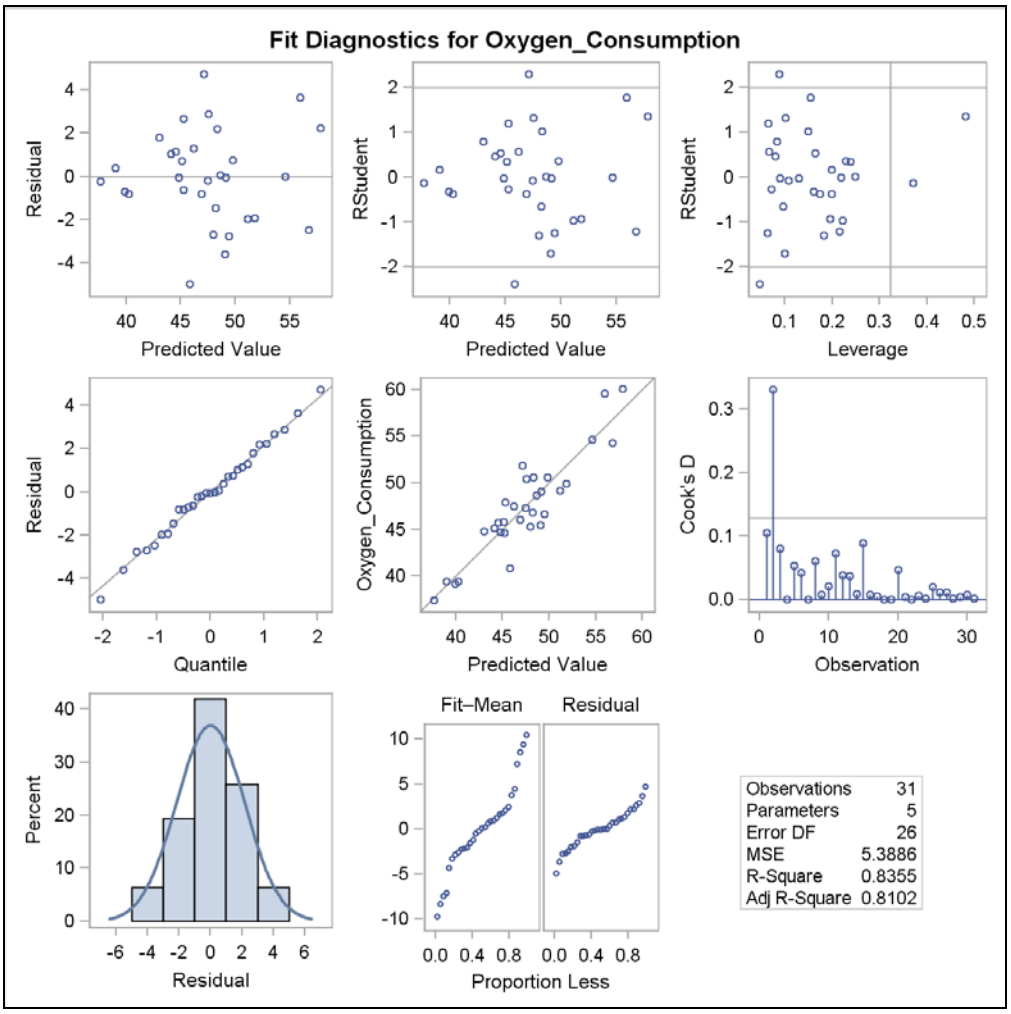
More diagnostic plots and plots are included by default, as well as a box plot for residuals.



A subset of the residual and diagnostic plots are produced in the Fit Diagnostics panel plot.

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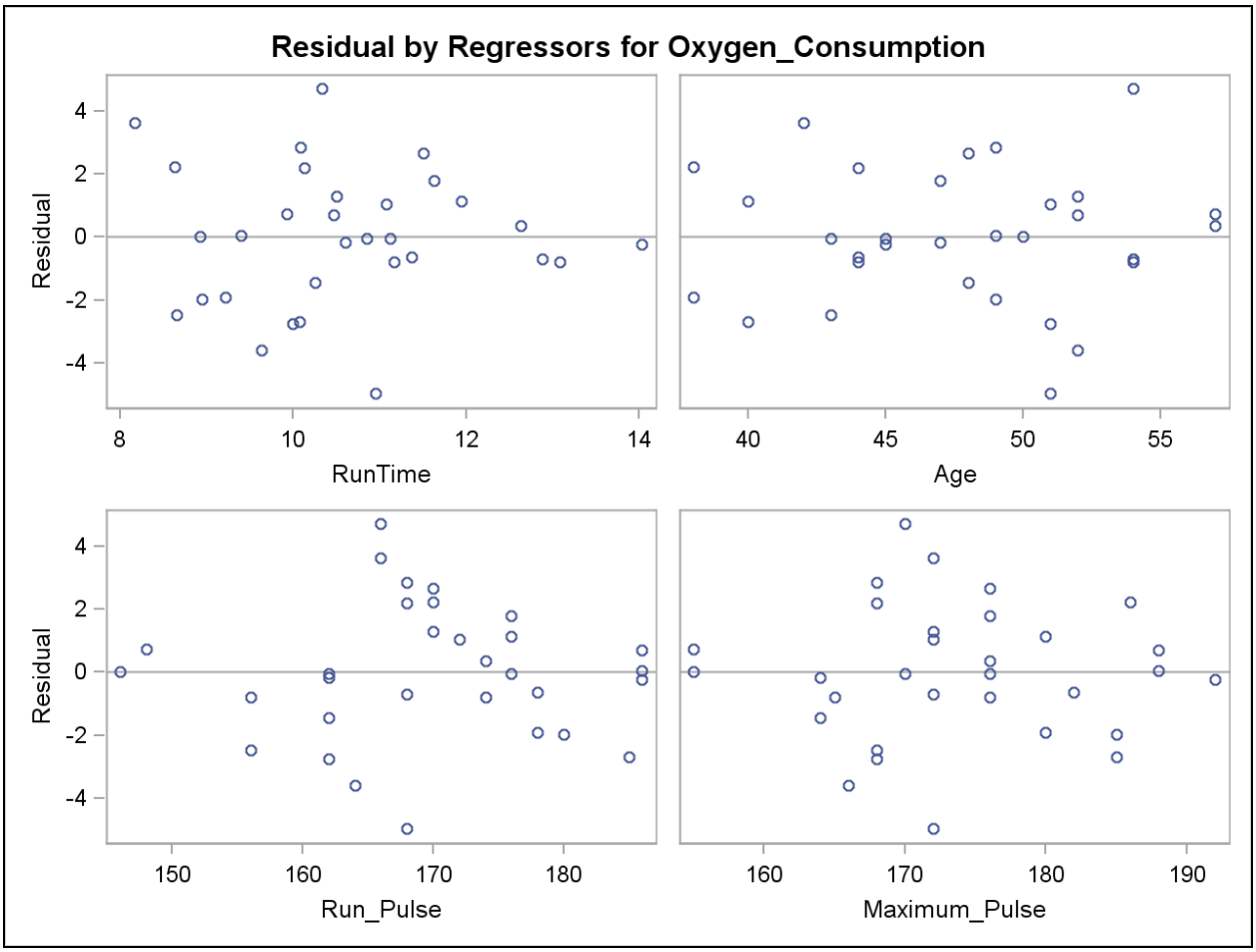

The plot of the residuals versus the values of the independent variables, **Runtime**, **Age**, **Run_Pulse**, and **Maximum_Pulse**, is shown above. They show no obvious trends or patterns in the residuals. Recall that independence of residual errors (no trends) is an assumption for linear regression, as is constant variance across all levels of all predictor variables.

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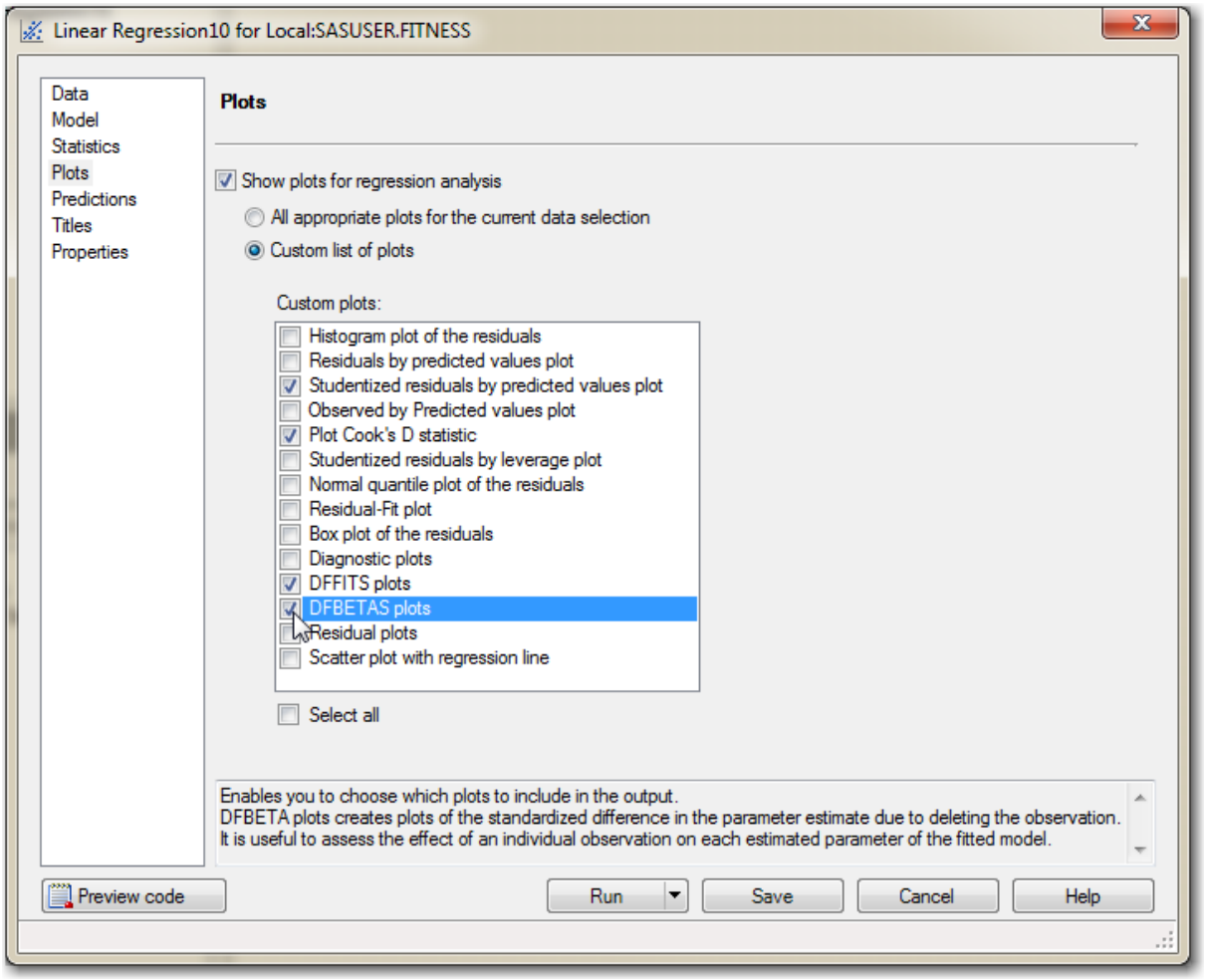
Looking for Influential Observations

Example: Generate only the RStudent, DFFITS, DFBETAS, and Cook's D influence plots for the Mallows C_p selected model. Save the statistics to an output data set.

1. Modify the previous task.
2. With **Plots** selected at the left, check the boxes shown checked below in the **Custom plots** area.

RSTUDENT residuals are referred to as **Studentized residuals** in the task windows.



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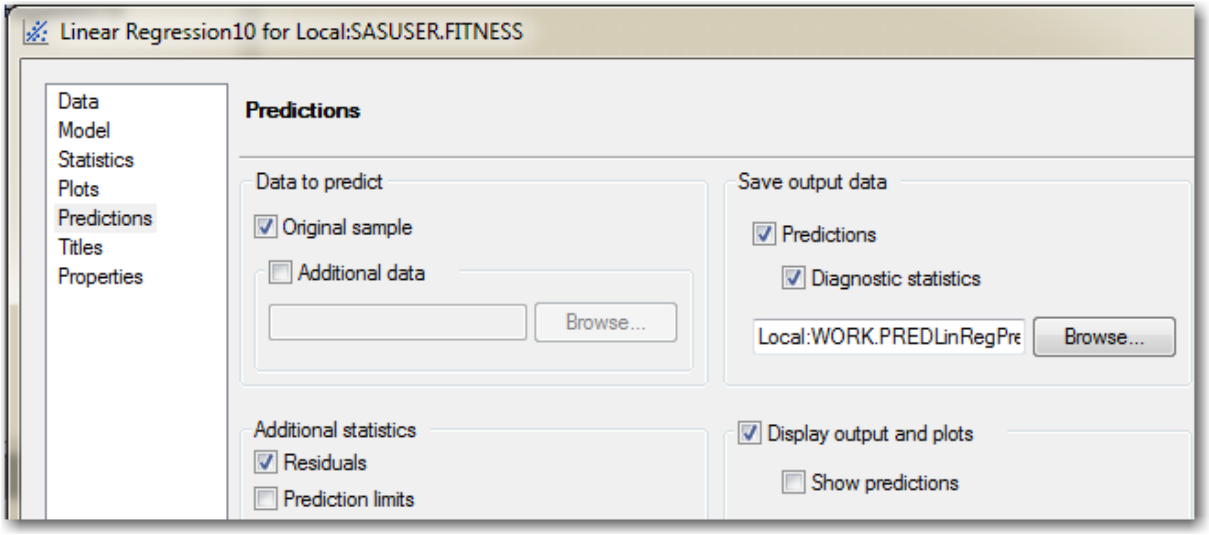
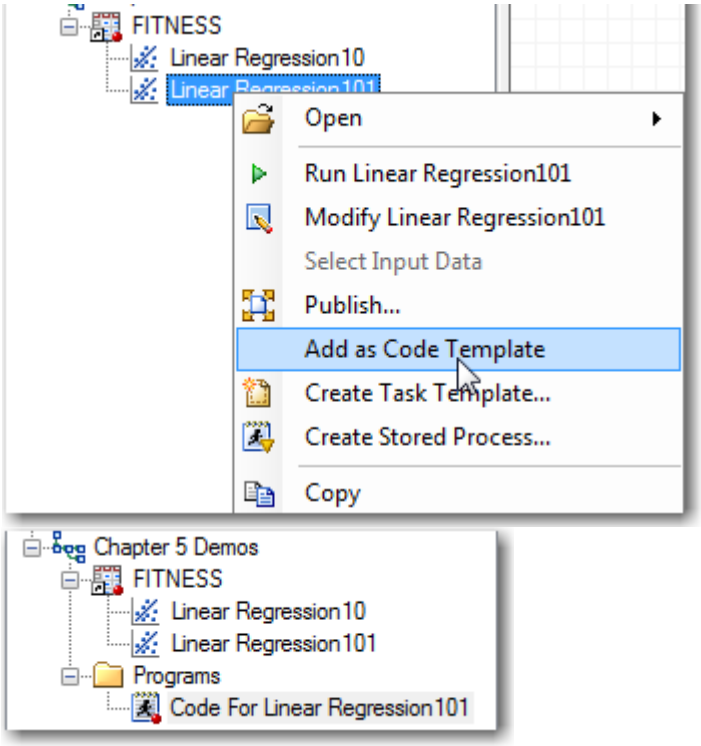
3. With **Predictions** selected at the left:
   a. Check the box for **Original sample** under **Data to predict**.
   b. Check **Predictions** and **Diagnostic statistics** under **Save output data**.
   c. Check the box for **Residuals** under **Additional statistics**.

   You can change the name and library of the data set where the diagnostic statistic variables will be stored by clicking **Browse** in the **Save output data** area.



4. Click **Save**. Do not replace the results from the previous run.
5. Right-click the previous task and select **Add as a Code Template**.



6. Double-click the node for the code in order to edit it and find the PROC REG section of the code.

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```
TITLE1 "Linear Regression Results";
FOOTNOTE;
FOOTNOTE1 "Generated by the SAS System (&_SASSERVERNAME, &SYSSCPL) on
%TRIM(%QSYSFUNC(DATE(), NLDATE20.)) at %TRIM(%SYSFUNC(TIME(),
TIMEAMPM12.))";
PROC REG DATA=WORK.SORTTempTableSorted
           PLOTS(ONLY)=RSTUDENTBYPREDICTED
           PLOTS(ONLY)=COOKSD
           PLOTS(ONLY)=DFFITS
           PLOTS(ONLY)=DFBETAS
      ;
      Linear_Regression_Model: MODEL Oxygen_Consumption = RunTime Age
Run_Pulse Maximum_Pulse
           /          SELECTION=NONE
      ;

      OUTPUT OUT=WORK.PREDLINREGPREDICTIONSFITNES_0001(LABEL=
"Linear regression predictions and statistics for SASUSER.FITNESS")
           PREDICTED=predicted_Oxygen_Consumption
           RESIDUAL=residual_Oxygen_Consumption
           STUDENT=student_Oxygen_Consumption
           RSTUDENT=rstudent_Oxygen_Consumption
           COOKD=cookd_Oxygen_Consumption
           DFFITS=dffits_Oxygen_Consumption
           H=h_Oxygen_Consumption
           STDI=stdi_Oxygen_Consumption
           STDP=stdp_Oxygen_Consumption
           STDR=stdr_Oxygen_Consumption ;
RUN;
QUIT;
```

7. Make the following changes:

   a. Add the option (LABEL) within the parentheses after the words PLOTS. These options will cause Enterprise Guide to label influential and other outliers in plots. If an ID statement is included, the value of that variable will appear on the plot. Otherwise, the observation number will act as the label.

```
PLOTS(ONLY LABEL)=RSTUDENTBYPREDICTED
PLOTS(ONLY LABEL)=COOKSD
PLOTS(ONLY LABEL)=DFFITS
PLOTS(ONLY LABEL)=DFBETAS
```

   b. Add the statement ID NAME; immediately above the OUTPUT statement.

```
ID NAME;
OUTPUT OUT=WORK.PREDLINREGPREDICTIONSFITNES_0001(LABEL=
"Linear regression predictions and statistics for SASUSER.FITNESS")
```

8. Click **Save** and then select **Save As** to name and locate the file.

9. Click **Run** above the code window.

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Output:

Number of Observations Read	31
Number of Observations Used	31

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	4	711.45087	177.86272	33.01	<.0001
Error	26	140.10368	5.38860		
Corrected Total	30	851.55455			

Root MSE	2.32134	**R-Square**	0.8355
Dependent Mean	47.37581	**Adj R-Sq**	0.8102
Coeff Var	4.89984		

Parameter Estimates					
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**
Intercept	1	97.16952	11.65703	8.34	<.0001
RunTime	1	-2.77576	0.34159	-8.13	<.0001
Age	1	-0.18903	0.09439	-2.00	0.0557
Run_Pulse	1	-0.34568	0.11820	-2.92	0.0071
Maximum_Pulse	1	0.27188	0.13438	2.02	0.0534

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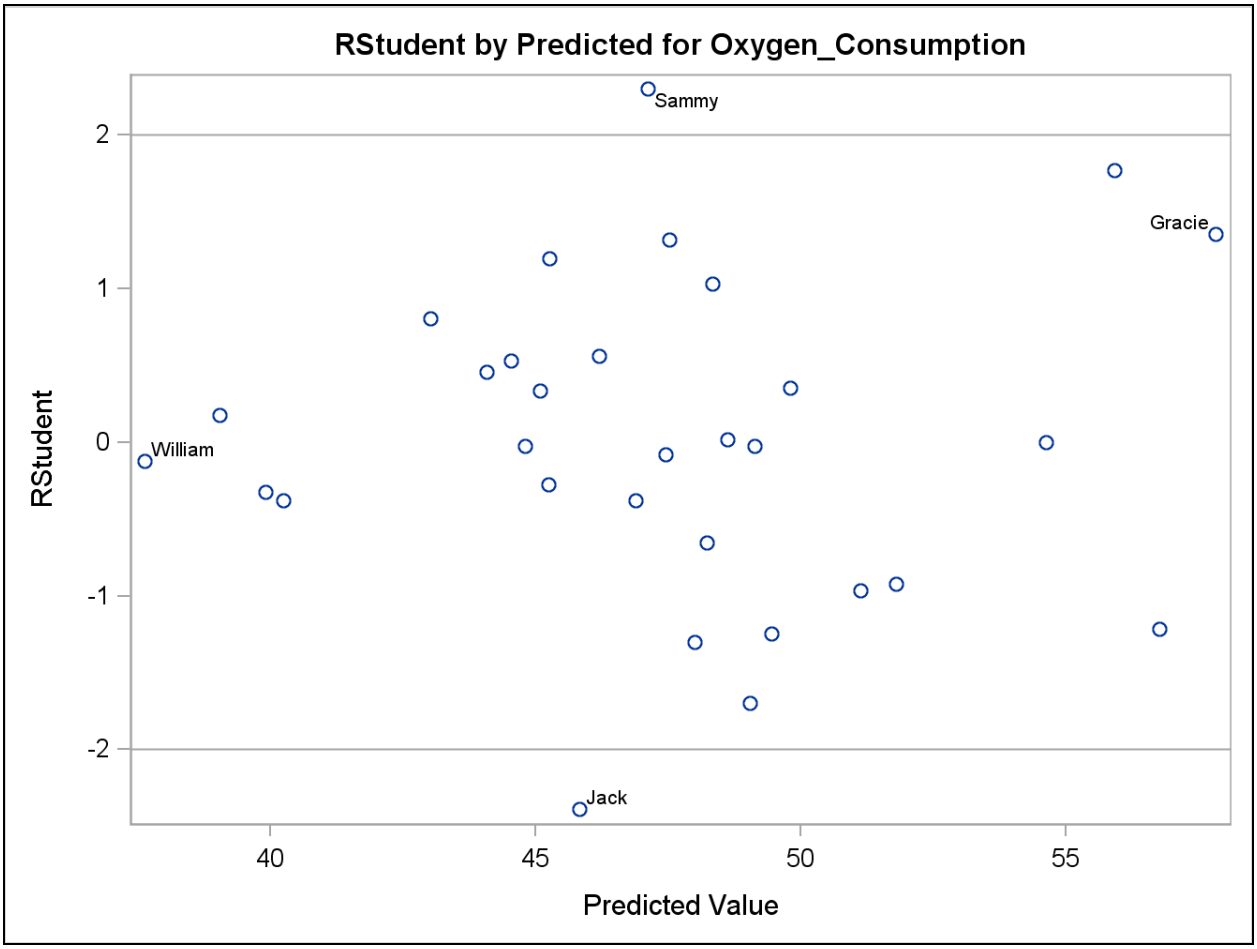
Partial PROC REG Output (Continued)



The RStudent plot shows two observations beyond two standard errors from the mean of 0. Those are identified as Sammy and Jack. Because you expect 5% of values to be beyond two standard errors from the mean (remember that RStudent residuals are assumed to be normally distributed), the fact that you have two that far outside the primary cluster gives no cause for concern (5% of 31 is 1.55 expected observations).

William and Gracie are also labeled in this plot because they have the most extreme predictor variable values. (Their leverage values are extreme.)

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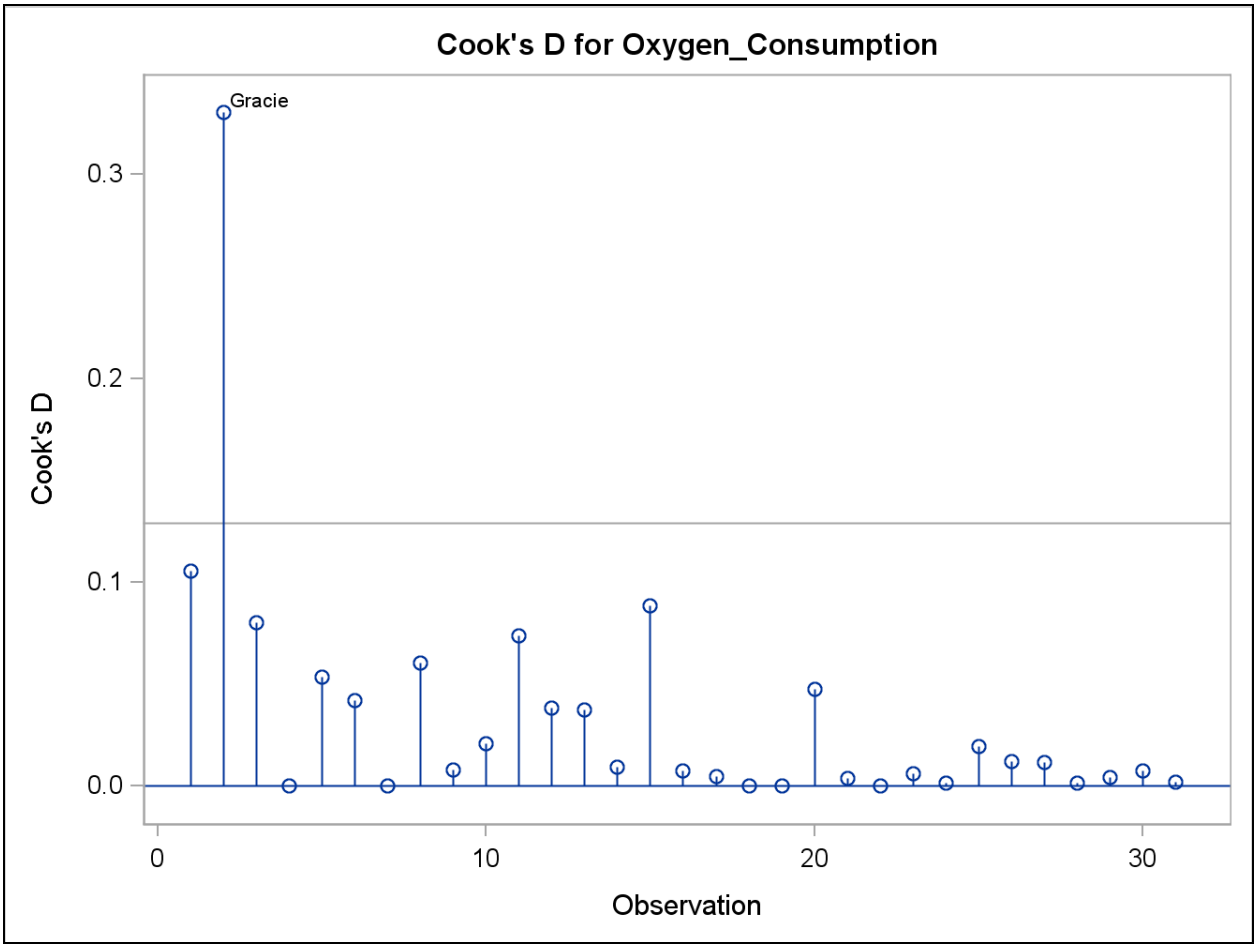

The Cook's *D* plot shows **`Gracie`** to be an influential point.

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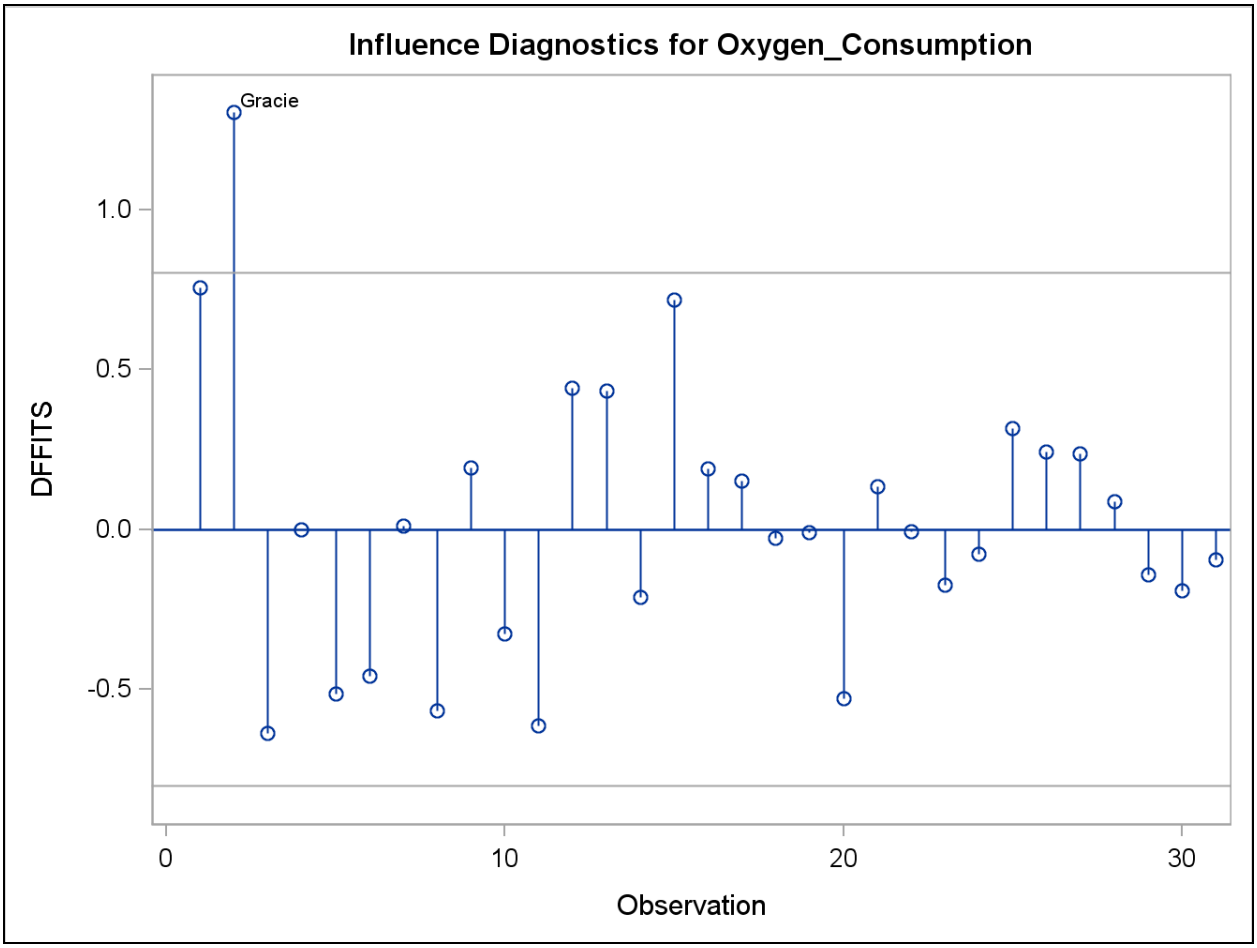

`Gracie` appears once again as an influential point based on her value on DFFITS.

At this point, it might be helpful to see which parameters **`Gracie`** might influence most. DFBETAS provides that information.

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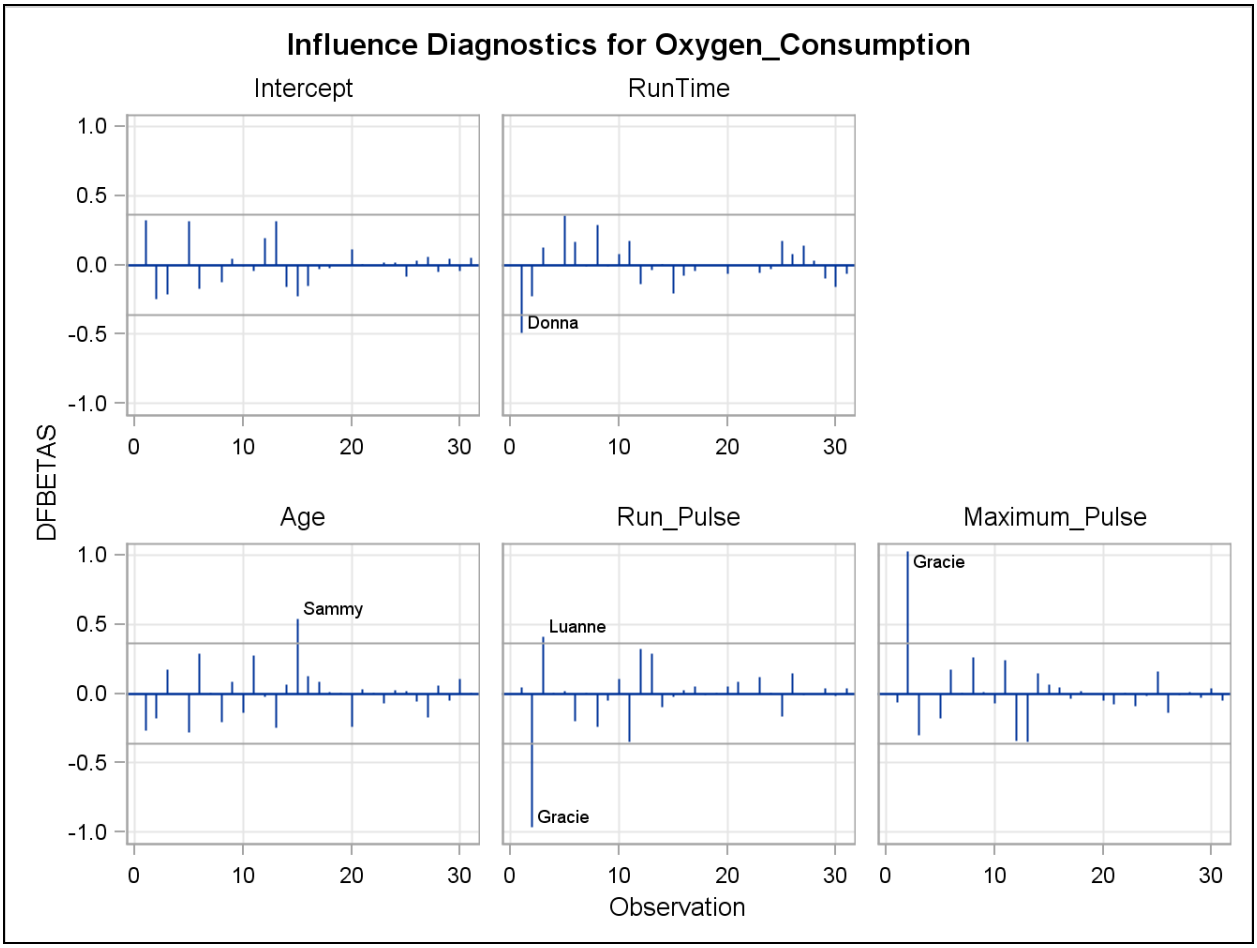

Apparently, **`Gracie`** is influential because of her effects on the estimates of both **Run_Pulse** and **Maximum_Pulse**.

Look at the output data set created from the previous program. Scroll all the way to the right of the file to find the variables containing values of Cook's D, RSTUDENT, and DFFITS.

Program* | Log | Output Data | Results

Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Export | Send To

	cookd_Oxygen_Consumption	h_Oxygen_Consumption	rstudent_Oxygen_Consumption	dffits_Oxygen_Consumption	stdi_Oxygen_Consumption
1	0.1054573125	0.1538247283	1.771779431	0.755427115	2.4934922464
2	0.3305135144	0.4824104315	1.3526483811	1.3058716835	2.8263265103
3	0.0799872241	0.2154643888	-1.217904396	-0.638255577	2.5592294255
4	9.7678684E-9	0.2192844436	-0.000408894	-0.000216705	2.5632479419
5	0.0534709771	0.2222803465	-0.965928366	-0.516397244	2.5663950872
6	0.0421367534	0.1964456637	-0.925773637	-0.457739518	2.5391279782

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Example of Collinearity

Example: Generate a regression model with **Oxygen_Consumption** as the dependent variable and **Performance**, **Runtime**, **Age**, **Weight**, **Run_Pulse**, **Rest_Pulse**, and **Maximum_Pulse** as the independent variables. Compare this model with the model from the previous section.

1. With the **Fitness** data set active, select **Tasks** ⇨ **Regression** ⇨ **Linear Regression**.
2. Drag **Oxygen_Consumption** to the dependent variable role and all other numeric variables to the explanatory variables role.
3. With **Plots** selected at the left, uncheck the box for **Show plots for regression analysis**.
4. Click **Run**.

Output

Number of Observations Read	31
Number of Observations Used	31

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	7	722.66124	103.23732	18.42	<.0001
Error	23	128.89331	5.60406		
Corrected Total	30	851.55455			

Root MSE	2.36729	**R-Square**	0.8486
Dependent Mean	47.37581	**Adj R-Sq**	0.8026
Coeff Var	4.99683		

Parameter Estimates					
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**
Intercept	1	131.78249	72.20754	1.83	0.0810
Performance	1	-0.12619	0.30097	-0.42	0.6789
RunTime	1	-3.86019	2.93659	-1.31	0.2016
Age	1	-0.46082	0.58660	-0.79	0.4401
Weight	1	-0.05812	0.06892	-0.84	0.4078
Run_Pulse	1	-0.36207	0.12324	-2.94	0.0074
Rest_Pulse	1	-0.01512	0.06817	-0.22	0.8264
Maximum_Pulse	1	0.30102	0.13981	2.15	0.0420

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For the full model, Model *F* is highly significant and the R-square is large (with adjusted R-square close to R-square). These statistics suggest that the model fits the data well.

However, when you examine the *p*-values of the parameters, only **Run_Pulse** and **Maximum_Pulse** are statistically significant.

Recall that the Mallows prediction model included **Runtime**. However, in the full model, this same variable is not statistically significant (*p*-value=0.2016). The *p*-value for **Age** changed from 0.0557 to 0.4401 between the models.

When you have a highly significant Model *F* but no (or few) highly significant terms, collinearity is a potential cause.

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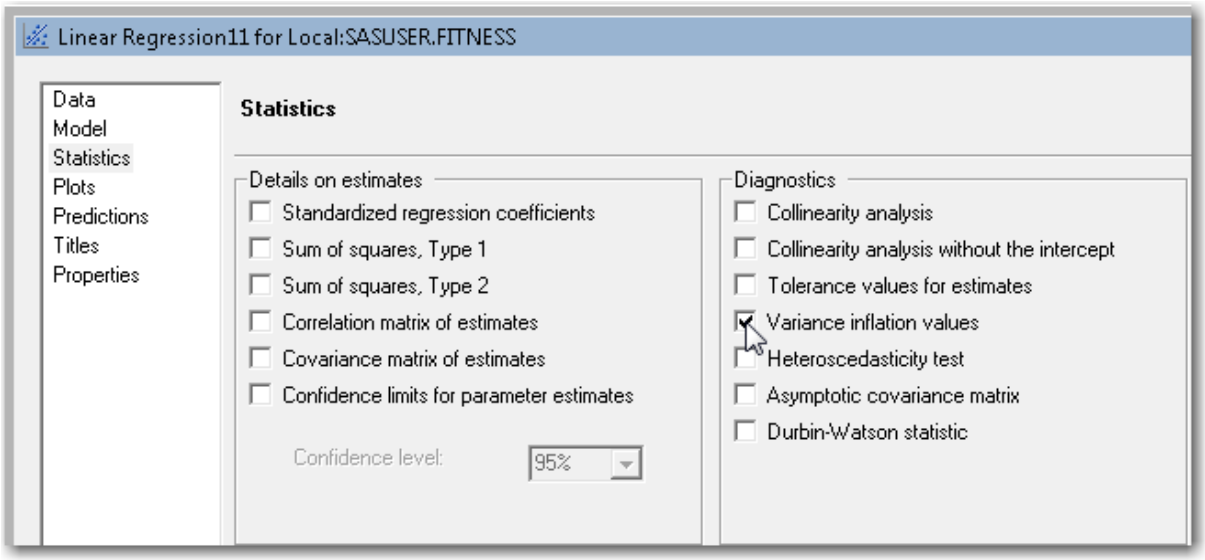
Collinearity Diagnostics

Example: Use the **Variance inflation values** option to assess the magnitude of the collinearity problem and identify the terms involved in the problem.

1. Modify the preceding Linear Regression task.
2. With **Statistics** selected at left, check the box for **Variance inflation values** under **Diagnostics**.



3. Click **Run**. Do not replace the results from the previous run.

Output

Number of Observations Read	31
Number of Observations Used	31

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	7	722.66124	103.23732	18.42	<.0001
Error	23	128.89331	5.60406		
Corrected Total	30	851.55455			

Root MSE	2.36729	**R-Square**	0.8486
Dependent Mean	47.37581	**Adj R-Sq**	0.8026
Coeff Var	4.99683		

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Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	131.78249	72.20754	1.83	0.0810	0
RunTime	1	-3.86019	2.93659	-1.31	0.2016	88.86251
Age	1	-0.46082	0.58660	-0.79	0.4401	51.01176
Weight	1	-0.05812	0.06892	-0.84	0.4078	1.76383
Run_Pulse	1	-0.36207	0.12324	-2.94	0.0074	8.54498
Rest_Pulse	1	-0.01512	0.06817	-0.22	0.8264	1.44425
Maximum_Pulse	1	0.30102	0.13981	2.15	0.0420	8.78755
Performance	1	-0.12619	0.30097	-0.42	0.6789	162.85399

Some of the variance inflation values are much larger than 10. A severe collinearity problem is present. At this point there are many ways to proceed. However, it is always a good idea to use some subject-matter expertise. For example, a quick conversation with the analyst and a view of the data-coding scheme revealed this bit of information.

Partial Code

```
data st192.fitness;
   input @1 Name $8. @10 Gender $1. @12 RunTime 5.2 @18 Age 2. @21
         Weight 5.2
         @27 Oxygen_Consumption 5.2 @33 Run_Pulse 3.
         @37 Rest_Pulse 2. @40 Maximum_Pulse 3.;
   Performance=260-round(10*runtime + 2*Age + 4*(Gender='F'));
   datalines;
...
run;
```

The variable **Performance** was not a measured variable. The researchers, on the basis of prior literature, created a summary variable, which is a weighted function of the three variables, **RunTime**, **Age**, and **Gender**. This is not at all an uncommon occurrence and illustrates an important point. If a summary variable is included in a model along with some or all of its composite measures, there is bound to be collinearity. In fact, this can be the source of great problems.

If the composite variable has meaning, it can be used as a stand-in measure for all three composite scores and you can remove the variables **RunTime** and **Age** from the analysis.

Summary measures have the disadvantage of losing some information about the individual variables. If this is of concern, then remove **Performance** from the analysis.

A decision was made to remove **Performance** from the analysis. Another check of collinearity is warranted.

4. Modify the preceding Linear Regression task.
5. With **Data** selected at left, remove **Performance** from the **Explanatory variables** list.
6. Click **Run**. Do not replace the results from the previous run.

Output

Number of Observations Read	31
Number of Observations Used	31

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Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	6	721.67605	120.27934	22.23	<.0001
Error	24	129.87851	5.41160		
Corrected Total	30	851.55455			

Root MSE	2.32629	**R-Square**	0.8475
Dependent Mean	47.37581	**Adj R-Sq**	0.8094
Coeff Var	4.91028		

Parameter Estimates						
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**	**Variance Inflation**
Intercept	1	101.96313	12.27174	8.31	<.0001	0
RunTime	1	-2.63994	0.38532	-6.85	<.0001	1.58432
Age	1	-0.21848	0.09850	-2.22	0.0363	1.48953
Weight	1	-0.07503	0.05492	-1.37	0.1845	1.15973
Run_Pulse	1	-0.36721	0.12050	-3.05	0.0055	8.46034
Rest_Pulse	1	-0.01952	0.06619	-0.29	0.7706	1.41004
Maximum_Pulse	1	0.30457	0.13714	2.22	0.0360	8.75535

The greatest VIF values are much smaller now. The variables **Maximum_Pulse** and **Run_Pulse** are also collinear, but for a natural reason. The pulse at the end of a run is highly likely to correlate with the maximum pulse during the run. You might be tempted to remove one variable from the model, but the small *p*-values for each indicate that this would adversely affect the model.

7. Modify the preceding Linear Regression task.
8. With **Data** selected at left, remove **Maximum_Pulse** from the **Explanatory variables** list.
9. Click **Run**. Do not replace the results from the previous run.

Output

Number of Observations Read	31
Number of Observations Used	31

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	5	694.98323	138.99665	22.19	<.0001
Error	25	156.57132	6.26285		
Corrected Total	30	851.55455			

Root MSE	2.50257	**R-Square**	0.8161
Dependent Mean	47.37581	**Adj R-Sq**	0.7794
Coeff Var	5.28238		

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Parameter Estimates						
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**	**Variance Inflation**
Intercept	1	115.46115	11.46893	10.07	<.0001	0
RunTime	1	-2.71594	0.41288	-6.58	<.0001	1.57183
Age	1	-0.27650	0.10217	-2.71	0.0121	1.38477
Weight	1	-0.05300	0.05811	-0.91	0.3704	1.12190
Run_Pulse	1	-0.12213	0.05207	-2.35	0.0272	1.36493
Rest_Pulse	1	-0.02485	0.07116	-0.35	0.7298	1.40819

With **Maximum_Pulse** removed, all of the VIF values are low, but the R square and Adjusted R square values were reduced and the *p*-value for **Run-Pulse** actually increased!

Even with collinearity still present in the model, it might be advisable to keep the previous model including **Maximum_Pulse**.

Collinearity can have a substantial effect on the outcome of a stepwise procedure for model selection. Because the significance of important variables can be masked by collinearity, the final model might not include very important variables. This is why it is advisable to deal with collinearity before using any automated model selection tool.

There are other approaches to dealing with collinearity. Two techniques *are ridge regression* and *principal components regression.* In addition, *recentering* the predictor variables can sometimes eliminate collinearity problems, especially in a polynomial regression and in ANCOVA models.

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